

Supplementing Patient Encounter Training for Preservice Nurses: Towards an AI×VR Approach

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Fig. Immersive VR clinic for AI-VSP encounters.

Motivation and problem context

- Standardized-patient (SP) encounters provide realism and formative feedback, but remain resource-intensive and difficult to scale consistently.
- Immersive VR can standardize exposure while maintaining embodied interaction; generative AI can enable open-ended, persona-consistent dialogue.
- This work investigates an AI×VR approach as a *supplement* to SP-based preparation.



Objective and research questions

- We developed an immersive VR clinic enabling preservice nurses to practice patient encounters with Artificial Intelligence-Driven Virtual SPs (AI-VSPs) via speech-based, open-ended conversation.
- The system incorporates varied patient personas and lightweight, transfer-oriented scaffolding (captions and cue sheets).
- Evaluation focuses on usability, self-efficacy change, transfer to SP performance, and perceived facilitators/barriers.



(a)



(b)

Fig. (a) A preservice nurse practices patient encounters with AI-VSPs in a spatial classroom, enabling full individual immersion.

Fig. (b) A preservice nurse completes a performance assessment with a SP at the nursing school.



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AI-VSP VR clinic: implementation and interaction

- Platform: Unity-based VR clinic deployed on Meta Quest 2; push-to-talk speech interaction.
- Conversational pipeline: ASR + LLM-based dialogue + TTS integrated via InWorld, supporting spoken interaction with an embodied avatar.
- Transfer-oriented supports: live captions for verification/repair and a floor cue sheet listing core history-taking prompts.

Push-to-talk: Hold the trigger to speak.

ASR: Automatic Speech Recognition.

LLM: Large Language Model.

TTS: Text to Speech



Case and persona design

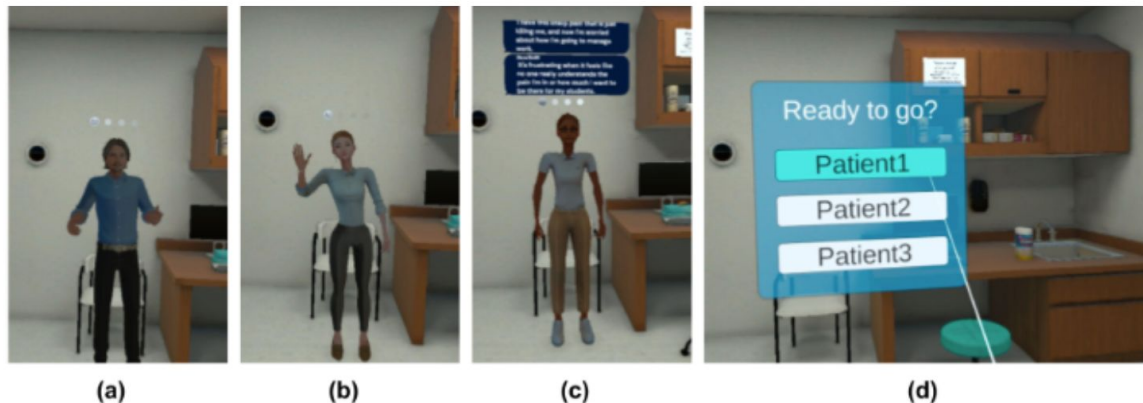


Fig. (a,b,c) AI-VSPs with introverted, ambiverted, and extroverted personas;

Fig. (d) floating patient-selection menu.

- Three AI-VSP personas (introverted, ambiverted, extroverted) were authored to elicit adaptation to patient communication style.
- Personas share an identical clinical case profile (acute low-back pain radiating to the right ankle), isolating persona effects from diagnostic content.



Study design and measures

- Mixed-methods, quasi-experimental design; AI-VSP practice was offered as an optional preparatory activity.
- Practice dose: ~30-minute AI-VSP session across three personas (experimental group).
- Assessment:
 - *Quantitative: both groups completed an SP encounter (~48 hrs); performance scored by two raters (15-item rubric, $\kappa = 0.70$).*
 - *Measured usability, self-efficacy, presence, immersion and engagement using standardized scales.*
 - *Qualitative: Identified key themes from interviews via thematic analysis.*

TABLE. RUBRIC FOR EVALUATING NURSING STUDENTS' PATIENT ENCOUNTER PERFORMANCE

Subdivision	Code	Short Description	Value
Clinical Assessment Items	C01	Explores patient's feelings	1
	C02	Identifies location of pain	1
	C03	Clarifies onset of pain	1
	C04	Assesses type/quality of pain	1
	C05	Assesses severity of pain	1
	C06	Reviews history of back problems	1
	C07	Identifies factors that relieve pain	1
	C08	Identifies factors that worsen pain	1
	C09	Assesses impact on function	1
	C10	Assesses emotional state	1
Greeting and Closure Items	G01	Provides appropriate greeting	1
	G02	Expresses courteous acknowledgment	1
	G03	Introduces self by name and role	1
	G04	Uses general well-being inquiry	1
	G05	Provides appropriate farewell	1



Findings

Quantitative findings

- **Usability** was high (SUS = 81.6/100), consistent with “excellent” usability.
- **Self-efficacy** increased following a single AI-VSP session ($3.74 \rightarrow 4.11$; $t(22)=3.82$, $p<0.001$).
- **SP encounter performance** did not differ between AI-VSP users and controls ($t(55) = -0.134$, $p = 0.894$).
- **AI-VSP performance** was not associated with engagement/presence/immersion, although experience measures were intercorrelated.

Qualitative findings

- **Engaging learning:** Participants reported strong engagement and a sense of presence attributed to the headset-based clinic and open-ended dialogue.
- **Educational benefits:** participants described improved redirection, elicitation of sparse information, and question sequencing, consistent with increased confidence.
- **Usability challenges:** latency/freezes, synthetic voice quality, and limited turn-taking control (e.g., overlong responses and weak interruptibility) increased workload.



Design and curricular implications

System design implications

- Constrain agent responses ($\approx 1-3$ sentences), enable reliable barge-in/interruptibility, and reduce ASR/TTS latency while improving prosody.
- Maintain a single-user clinic with minimal interface overhead; use concise, rubric-aligned scaffolds (captions + cue sheets) to support coverage without scripting.

Curricular integration implications

- Position AI-VSP sessions as low-stakes, on-demand warm-ups prior to SP encounters; distribute multiple short “micro-sessions” across a module.



Fig. Immersive VR clinic for AI-VSP encounters.



Conclusions

- An AI-VSP VR clinic can function as a usable and confidence-building supplement to SP encounter preparation.
- Observed outcomes indicate increased self-efficacy with performance parity relative to traditional preparation, supporting a supplemental (rather than replacement) role.
- Interaction mechanics (latency, verbosity, and interruptibility) emerge as primary determinants of educational utility and perceived realism.



Thank You!

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